PREDICTIVE ANALYTICS · TIER C · INTERNAL

Predictive Analytics Run — Proposal Input Form

Information to be collected for the monthly operation of the models in production — drift and performance monitoring, retraining, monthly report — Run (C). Sections follow the Proposal Set codes (P1, P2, P5, P8). Once the form is completed and returned, the proposal text (file 4c) is produced. Orange tags mark drafts: statements with the AX tag are to be verified by the Analytics (AX) team (use cases, data, metrics, effort, bands, team, references, acceptance), statements with the GÜNER tag by Güner Hacıoğlu (cloud / platform items, the MACFit and DeFacto references).

Rows marked ● are BLOCKERS — the proposal cannot be written without them and they cannot be filled with assumptions.

If rows marked ○ remain missing, they are written into the proposal as assumptions and go to the Assumptions and Out of Scope section.

The "WHERE IT FILLS IN THE PROPOSAL" column of each row names the section or variable field of the proposal template (file 4c). This form uses the Proposal Set codes P1, P2, P5, P8; the A·B·C mark in each code heading shows the tiers in which the code is asked.

If the answer to a question is "we don't know", that is also an answer — it is often the first finding and goes into the proposal as an assumption.

LIMIT OF TIER C: C is the monthly operation of the models in production; it is priced by monthly capacity hours. A new use case is not built from C capacity; a new use case is a separate A or B proposal.

ACTIVITIES: drift and performance monitoring; retraining schedule (periodic or conditional); monthly report; small improvements from the monthly capacity. «AX DOĞRULASIN»

CAPACITY AND TERM: monthly S 40 h · M 80 h · L 160 h; minimum term 12 months (draft); unused hours are not carried over to the next month; work above capacity at the «hour rate». «AX DOĞRULASIN»

OUT OF SCOPE: a new use case or a model redesign with new data sources; operation of the data platform and source system errors; execution of campaigns, price changes or operational actions. «AX DOĞRULASIN»

CLOUD: on Google Cloud, cloud consumption (BigQuery, Vertex AI) is not included in the C fee and is invoiced separately; tracking quota, cost and service changes on the Vertex AI / BigQuery side is a C activity. «GÜNER DOĞRULASIN»

PLATFORM: the delivery platform can be Google Cloud (BigQuery, BigQuery ML, Vertex AI), Dataiku, Azure ML or an open-source stack. The lifecycle and the tiers do not change with the platform; the platform choice is a scope item («PLATFORM»).

NEIGHBOURING PACKAGES AND SEPARATE ASSETS: AI Readiness Assessment measures readiness, this package builds the model; ETL Modernization on Google Cloud and Data Modernization move the data platform; AI Governance Readiness covers model governance. The Analytic CoE Enablement Program, the MLOps / AI Factory platform and the analytics platform migration to Dataiku (SAS / R → Dataiku, Pentaho → Dataiku) are separate candidate assets; RAG / LLM work belongs to Gemini Agent Kit / Grounded Intelligence.

P1 · A·B·C · IDENTITY — Organisation and Proposal Details

	TO ASK	WHERE IT FILLS IN THE PROPOSAL	ANSWER
●	**Full legal name of the organisation and the short name used in the text; single company or group with subsidiaries**	Cover «CUSTOMER NAME», text «CUSTOMER»	
○	Legal entities, brands, plants or group companies in scope — names; for a single company the row is deleted	Cover «GROUP COMPANIES», Scope	
○	Sector and reason for the request — for the one-paragraph context	Purpose «CUSTOMER CONTEXT»	
○	Is there an RFP, specification or requirements list; its number, response format and deadline	Structure and response format of the proposal	
○	Proposal addressee — name, title, unit; proposal date and expected decision date	Cover «dd Month yyyy», Commercial Terms «proposal date + 1 month»	
○	Procurement process, approved budget and expected invoice payment term; is monthly invoicing accepted	Effort and Fees, Commercial Terms «NUMBER» days payment term	
○	If the platform is Google Cloud, where is cloud consumption (BigQuery, Vertex AI) bought today — directly, through Komtaş, through another partner or reseller; is there a Dataiku or Azure licence	internal: Commercial frame	

	TO ASK	WHERE IT FILLS IN THE PROPOSAL	ANSWER
○	Are there competitors, and who (another partner, a software product, the customer's own data science team)	Emphasis and section weights	

P2 · A·B·C · DECISION — Use Case, Business Goal, Platform, Tier and Sponsor

> For models Komtaş did not build, an onboarding review is carried out before C starts; effort «NUMBER» pd, written as a separate item. «AX DOĞRULASIN»

	TO ASK	WHERE IT FILLS IN THE PROPOSAL	ANSWER
●	**Source of the operations need — models Komtaş took into production in B, the customer's own models, or models taken over from another partner**	Scope «PRODUCTION REF» / «PILOT REF», Purpose, Assumptions and Out of Scope	
○	Use cases to operate — family and use case (from the Use-Case Catalogue)	ANNEX-1 — Family Pages (selected family block), Driver 1	
○	Which tiers are wanted — C only; is there interest in a new use case (A or B)	Next Step, Service Selection Check	
●	**Delivery platform — Google Cloud (BigQuery, BigQuery ML, Vertex AI), Dataiku, Azure ML or an open-source stack; existing environment, licences and the customer's platform preference**	Scope «PLATFORM», Approach	
○	Requested start date — end of the B handover or end of the existing support contract	Activities and Timeline, Commercial Terms	
○	Who operates the models today; is there an existing support contract	Customer Responsibilities, Assumptions and Out of Scope	

Use-Case Catalogue — Five Families

The unit sold is one use case: Pilot one use case, Production per use case, Run monthly. The family sets the block that stays in ANNEX-1 of the proposal. The catalogue is compiled from the frameworks in the AX deck (table as a whole). «AX DOĞRULASIN»

#	FAMILY (PROPOSAL BLOCK)	USE CASES	SECTOR AND ENTRY VIEW
1	Retention & Churn («FAMILY — RETENTION & CHURN»)	Path to Churn; identifying "balance" churners; asset churn / segment downgrade (bank); subscription churn (fitness / subscription); customer satisfaction index	Finance — Retention; fitness / subscription; retail
2	Demand & Supply («FAMILY — DEMAND & SUPPLY»)	Demand forecasting (retail, manufacturing); inventory productivity; visitor and sales prediction (shopping malls); schedule optimization	Retail; manufacturing — Supply Chain; shopping mall management
3	Pricing & Personalization («FAMILY — PRICING & PERSONALIZATION»)	Next best offer; product bundling; recommendation; customer lifetime value; deposit pricing; credit pricing; list price and markdown optimization; dynamic pricing; customer segmentation; customer contact policy; onboarding analytics	Finance — Acquisition & Onboarding, Growth, Pricing & Personalization; retail; e-commerce; B2B spare parts
4	Risk & Collections («FAMILY — RISK & COLLECTIONS»)	Collection analytics / optimization; credit risk model enhancement (incl. graph analytics on relational data); dynamic client limit; risk analytics	Finance — Risk
5	Industrial Operations («FAMILY — INDUSTRIAL OPERATIONS»)	Energy optimization (HVAC, furnace); predictive quality / quality classification; production yield optimization; predictive maintenance; process unit prediction (refinery); fixed asset utilization — largely ML-guided optimization	Manufacturing — Resource Management, Operational Excellence; pharmaceuticals; refining. References mostly in glass manufacturing; further sectors on request.

P5 · B·C · FREQUENCY — Prediction Frequency and Timing

	TO ASK	WHERE IT FILLS IN THE PROPOSAL	ANSWER
●	**Prediction / scoring frequency per use case — monthly, weekly, daily or intra-day**	Driver 4, Scope «FREQUENCY», Effort and Fees	
○	When the scores / forecasts must be ready — e.g. Monday 07:00 or every 4 hours; business impact of a delay	Deliverables and Acceptance, Monthly Cycle	
○	Need for real-time (online) scoring — out of scope unless specified; batch scoring is the basis	Assumptions and Out of Scope	

P8 · C · OPERATIONS — Models, Retraining, Capacity and Term

> The price of C is the monthly capacity in hours. Capacity is set by the band from the use case, data source, volume, frequency and integration drivers.

	TO ASK	WHERE IT FILLS IN THE PROPOSAL	ANSWER
●	**Number of models to operate, their use cases and platform (BigQuery ML, Vertex AI, Dataiku, Azure ML, open-source, other)**	Scope «PLATFORM», Driver 1	
●	**Expected retraining frequency — monthly, quarterly or conditional (triggered by drift / performance)** «AX DOĞRULASIN»	Retraining Schedule, Effort and Fees	
○	Drift / performance thresholds «NUMBER» that trigger retraining	Drift and Performance Thresholds	
○	Number of data sources feeding the models and of target systems receiving the scores	Drivers 2 and 5, Scope	
○	Volume — number of customers / accounts, time series, sensor tags	Driver 3	
○	Monthly capacity — S / M / L (from the band estimate); expected small improvements	Monthly Capacity, Effort and Fees	
○	Term — minimum 12 months is a draft	Commercial Terms	
○	Recipients and content of the monthly report — technical metrics, drift, business metric	Monthly Cycle, Deliverables and Acceptance	
○	The customer's own team and the split of responsibilities (RACI); owner of data pipeline failures; expected response time for a failed scoring run, and the ticketing tool	Customer Responsibilities, Support Hours and Priorities «SUPPORT HOURS»	

Band Estimate — Tier C

Completed after the answers are collected. In C the band gives the monthly capacity. The band is the highest band reached by at least three drivers. A driver reaches a band when its value falls in that band or a higher one; if three drivers do not reach M or L, the band is S. In Industrial Operations volume is measured by the number of sensor tags; thresholds «NUMBER». «AX DOĞRULASIN»

DRAFT — no recorded effort; bands and efforts are to be updated with the Analytics (AX) team's data. The effort of the AX references and of the MACFit / DeFacto work is unknown. Komtaş effort is not written into the proposal before the delivery team confirms it.

#	DRIVER	SMALL	MEDIUM	LARGE	RESULTING BAND
1	Use cases (P2 / P8)	1	2	≥ 3	
2	Data sources (P8)	≤ 3	4 – 8	> 8	
3	Volume: customers / time series (P8)	≤ 100k / ≤ 10k	100k – 1M / 10k – 100k	> 1M / > 100k	
4	Prediction frequency (P5)	monthly	weekly	daily / intra-day	

#	DRIVER	SMALL	MEDIUM	LARGE	RESULTING BAND
5	Systems to integrate (P8)	1	2 – 3	> 3	
	RESULT — monthly capacity «AX DOĞRULASIN»	40 h / month	80 h / month	160 h / month	

TEAM — Run lead + ML Engineer. «AX DOĞRULASIN»

CONDITIONS — unused hours are not carried over to the next month; work above capacity is invoiced at the «hour rate» with the customer's approval; the monthly report shows how the capacity was used.

ARCHITECTURE — Google Cloud: sources → BigQuery → BigQuery ML or Vertex AI (training, model registry, pipelines, batch prediction) → score / forecast tables → export to business systems → monitoring (drift, performance) → Looker reports. «GÜNER DOĞRULASIN» On the other platforms the same steps are built with Dataiku (flow, model training, deployment, monitoring), Azure ML or an open-source stack (e.g. MLflow, Airflow). «AX DOĞRULASIN»

Fields Filled by the Proposal Team

These fields are not asked of the customer; SAT and sales fill them from the band, the effort calculation, the payment pattern and approved lists.

#	FIELD	WHERE IN THE PROPOSAL	HOW IT IS SET	VALUE
1	KOMTAS-«CUSTOMER ABBREVIATION»-PRA-C-«YEAR»-«SEQUENCE»	Cover — proposal number	Proposal register sequence	
2	«dd Month yyyy» · «proposal date + 1 month» · «price validity date»	Cover, Commercial Terms	Send date + 1 month (P1); sales sets the price validity date	
3	«CUSTOMER CONTEXT»	Purpose	P1 and P2 answers	
4	«FAMILY — RETENTION & CHURN» / «FAMILY — DEMAND & SUPPLY» / «FAMILY — PRICING & PERSONALIZATION» / «FAMILY — RISK & COLLECTIONS» / «FAMILY — INDUSTRIAL OPERATIONS»	ANNEX-1 — Family Pages, Scope	P2 / P8 — the family blocks of the operated use cases stay, the others are deleted	
5	«USE CASE» · «PLATFORM»	Scope, Approach	P2 — name as in the Use-Case Catalogue; platform from the P2 answer	
6	«PRODUCTION REF» · «PILOT REF»	Scope	P2 — source of the models; deleted if Komtaş did not build them	
7	«BAND» · «NUMBER» h / month · «NUMBER» months	Effort and Fees, Commercial Terms	Band estimate, P8	
8	Retraining frequency · drift thresholds «NUMBER» · monthly report recipients	Retraining Schedule, Drift and Performance Thresholds, Monthly Cycle	P8 answers	
9	«monthly fee» · «hour rate» · «amount» · «day rate» · onboarding «fee»	Effort and Fees	Monthly capacity × hour rate; work above capacity at the hour rate; change requests at the day rate; onboarding review as a separate item (P2)	
10	Next step items	Next Step	Follow-on Interest table; no margin figures	
11	ANNEX-2 references table · «RESULT»	ANNEX-2 — About Komtaş	From the References and Results table below, the rows that fit the selected family; the naming and result rule is in the note of the table.	
12	«full name» · «title» · «e-mail / phone» · «CITY»	Commercial Terms	Komtaş commercial and technical contact; city for free on-site meetings	

References and Results — Source of ANNEX-2

Source: the AX “References & Services 2026” deck; rows 10–11 from Güner Hacıoğlu. Results are written only from this table; where no result is known, «RESULT» stays. “Named” rows may appear as references in customer files; each reference table / list in a customer file carries one AX verification tag. **«AX DOĞRULASIN»**

#	USE CASE	FAMILY	REFERENCE (INTERNAL)	MEASURED RESULT	IN THE PROPOSAL
1	Collection optimization	Risk & Collections	Albaraka Banking Group, TEB	3–7% reduction in transition into the first delinquency cycle; Gini 0.86; explainable AI; contact policy optimised	named
2	Credit risk model enhancement	Risk & Collections	QNB Finansbank	+4.5% credit approval rate at the same NPL ratio; Gini 0.78–0.81; graph analytics	named
3	Asset churn prediction	Retention & Churn	Odeabank	11% lower segment downgrade in the test group vs the control group	named
4	Energy optimization in HVAC	Industrial Operations	Eczacıbaşı Monrol	2.3% lower energy consumption in test units vs control	named
5	Furnace energy, furnace temperature control, glass quality classification, production yield	Industrial Operations	Şişecam	«RESULT»	named
6	List price and markdown optimization	Pricing & Personalization	a spare-parts seller	+4% revenue at the same conversion in the test group; price elasticity for 30,000+ SKUs; 12 A/B scenarios	unnamed
7	Demand prediction	Demand & Supply	a tyre manufacturer	7% lower stock holding cost for 7 tyre types	unnamed
8	Visitor and sales prediction	Demand & Supply	a shopping mall management company	Forecasts for 350+ stores in 2 malls and 1.6 million visitors	unnamed
9	Process unit prediction (CCR)	Industrial Operations	an integrated oil / petrochemical group	Qualitative (more stable, less error-prone operation)	unnamed
10	Subscription churn	Retention & Churn	MACFit	«RESULT»; POC or live unknown	anonymous
11	Demand forecasting (retail)	Demand & Supply	DeFacto	«RESULT»; POC or live unknown	anonymous
12	Recommendation	Pricing & Personalization	none	—	—

“Unnamed” rows stay unnamed as in the deck. MACFit and DeFacto are written anonymously in customer files (“a fitness / subscription company”, “a retail company”). **«GÜNER DOĞRULASIN»**

Service Selection Check

Before filling in the form, make sure that Tier C is the right service. One request can match more than one row.

STATUS	RIGHT SERVICE
It is unclear whether the data is ready for a model; readiness is to be measured first	AI Readiness Assessment
The value of a model for one catalogue use case is to be shown on the customer’s data	A — Pilot
A validated model from the Pilot or an equivalent study exists; production is wanted	B — Production
The model is in production; monitoring, retraining and a monthly report are wanted	C — Run
Another use case is wanted in addition to a use case with a completed Pilot	A — Pilot for the new use case; B if it is validated
The data sits in on-prem ETL (e.g. ODI) or a DWH; it needs to move to a new platform	ETL Modernization on Google Cloud / Data Modernization
The organisation’s analytics team, use-case roadmap and governance are to be set up together	Analytic CoE Enablement Program (separate asset)

STATUS	RIGHT SERVICE
An enterprise MLOps / AI Factory platform for many models is wanted	MLOps / AI Factory platform (separate asset)
Existing SAS / R models or Pentaho ETLs are to be moved to Dataiku	Analytics platform migration to Dataiku (separate asset)
The request is a generative AI agent, assistant or RAG / LLM	Gemini Agent Kit / Grounded Intelligence
The request is model governance, risk classification, policy	AI Governance Readiness
The customer wants a guaranteed business result or real-time scoring	No guarantee is given; real-time is clarified separately

Follow-on Interest

Follow-on work the customer may be interested in; feeds the Next Step section of the proposal. No margin figures appear in the proposal.

#	FOLLOW-ON WORK	RELATED TRIGGER	INTEREST (YES / NO / LATER)	NOTE
1	Additional use case — from the same or another family	Other business questions — the five catalogue families		
2	Data platform modernisation (ODI → GCP style) — ETL Modernization on Google Cloud, Data Modernization	Data in on-prem ETL / DWH		
3	MLOps / AI Factory platform; Analytic CoE Enablement Program (separate assets)	Number of models is growing; the organisation wants to build its own analytics team		
4	AI Governance Readiness	Governance of models in production		
5	Cloud consumption — BigQuery, Vertex AI on Google Cloud (through Komtaş)	Training and scoring workloads		

COMMERCIAL FRAME — INTERNAL ONLY. This box, its rates and the customer names in it are not written into the customer deck or the proposal.

The Google consumption margin is thin: Premier base 12%, 13–15% with incentives, ~5–6% in large deals; Azure 3%. Value comes from the use cases and C Run (recurring revenue). On Google Cloud the work also drives BigQuery / Vertex AI consumption and opens data platform work.

Evidence: the Analytics (AX) team's "References & Services 2026" deck — references with measured results (Albaraka, TEB, QNB Finansbank, Odeabank, Eczacıbaşı Monrol) and references without a known result (Şişecam). Whether the MACFit (subscription churn) and DeFacto (retail demand forecasting) work brought by Güner Hacıoğlu was POC or live, and its results, are unknown; these two customers stay anonymous in the proposal and the deck. There is no field reference for recommendation. Industrial Operations references are mostly in glass manufacturing; further sectors on request.

Cross-sell: at MACFit there is also an ODI → Google Cloud data platform opportunity (ETL Modernization on Google Cloud). When the data sits in on-prem ETL, migration and model work can be planned together.

Blocker Checklist

These rows must be answered before proposal writing starts.

SECTION	BLOCKER QUESTION	ANSWERED?
P1	Full legal name of the organisation and the short name used in the text; single company or group with subsidiaries	
P2	Source of the operations need — models Komtaş took into production in B, the customer's own models, or models taken over from another partner	
P2	Delivery platform — Google Cloud (BigQuery, BigQuery ML, Vertex AI), Dataiku, Azure ML or an open-source stack; existing environment, licences and the customer's platform preference	
P5	Prediction / scoring frequency per use case — monthly, weekly, daily or intra-day	

SECTION	BLOCKER QUESTION	ANSWERED?
P8	Number of models to operate, their use cases and platform (BigQuery ML, Vertex AI, Dataiku, Azure ML, open-source, other)	
P8	Expected retraining frequency — monthly, quarterly or conditional (triggered by drift / performance)	

IF ANY BLOCKER IS MISSING — the proposal is not written. If it must be written, each missing item is written into the Assumptions and Out of Scope section of the proposal as an assumption, and it is stated that the scope will be reassessed if that assumption does not hold.

The first two rows of P8 have a special place: without the number of models the band and capacity cannot be written, and without the retraining frequency the service conditions cannot be written.

Blocking conditions and their effect on the proposal

#	CONDITION	EFFECT ON THE PROPOSAL	ACTION	STATUS
1	Models not built by Komtaş	An onboarding review precedes C	Clarify the source in P2	
2	Insufficient new data for retraining	The retraining schedule cannot be committed	Clarify with the data owner in P8	
3	Target definition (label) changed or not documented	Performance monitoring loses its meaning	Confirm the definition in P8	
4	No control group possible	The monthly report shows technical metrics only	Clarify in P8	
5	No KVKK approval for continued processing (personal data)	C cannot start	Clarify the approval with the customer	
6	Retraining frequency and drift thresholds unknown	Capacity and service conditions cannot be written	Agree with the customer in P8	